아동 능력 형성의 추정

Estimating Child Skill Formation: A Korean Perspective

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대전제 및 개요

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2. Identifying Skill formation
3. 한국 컨텍스트 관점에서

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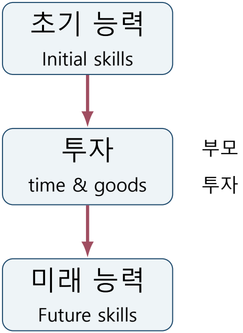
부모의 투자는 아동의 능력 형성에 어떻게 기여하는가?

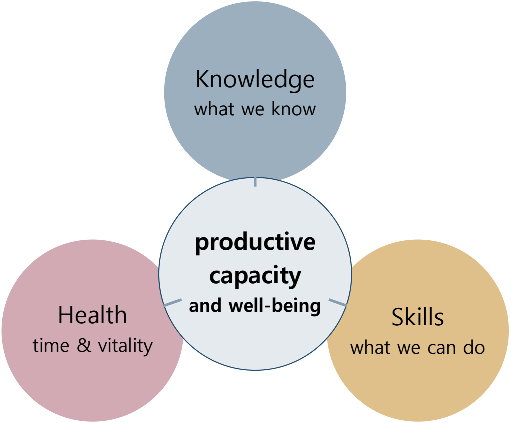
Why Child Skill Formation?

Parental Investment와 사교육

- Child Skill Formation 리터러쳐의 초점은 0-12세의 투자
- 미국 및 유럽에서는 Child Care에 더 포커스
- 동아시아는 Child Care 및 사교육이 중요한 쟁점
- 사교육을 포함한 한국 부모의 투자는 Skill formation에 얼마나 영향을 미칠까?



Human capital is multidimensional



지식, 건강, 기술과 능력은 서로 구분되지만, 함께 개인의 생산능력과 삶의 성과를 형성한다. Becker (1964)

Why does human capital matter?

Becker's organizing idea

"The concept of investment in human capital simply organizes and stresses these basic truths." (Becker, 1964)

1

Input (투입)

시간, 재화, 환경

2

Formation (형성)

현재 능력의 변화

3

Outcomes (성과)

교육, 소득, 건강, 삶의 질

Basic Idea: The production technology

Skill Stock + Investment $\rightarrow$ Future Skill

$$\theta_{t+1} = f(\theta_t, I_t, \varepsilon_t)$$

능력(Skill)

- θ_t: Skill at time t. 시점 t의 인지적 (Cognitive), 비인지적 (Non-cognitive) 능력
- 잠재적 능력 $\rightarrow$ 여러가지 measure Z_j를 통해 나타남 $(j = 1, 2, ..., J)$

투자(Investment)

- I_t: 시점 t의 가계,학교의 투자
- (특히 부모의) 시간, 교육지출, 보육, 공교육

Cunha and Heckman (2007); Cunha et al. (2010)

쟁점

1. **Who benefits most from investment?**

$$\frac{\partial^2 \theta_{\mathrm{t+1}}}{\partial \theta_{\mathrm{t}} \partial \mathrm{I}_{\mathrm{t}}}$$

   능력 수준에 따라 투자의 효과가 달라지는가?

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2. **When is investment most productive?**

$$\frac{\partial \theta_T}{\partial \mathrm{I}_t}$$

   어느 시점의 투자가 최종 능력에 가장 큰 영향을 미치는가?

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3. **How do early and later investments interact?**

$$\frac{\partial^2 \theta_T}{\partial \mathrm{I}_t \partial \mathrm{I}_{t+1}}$$

   초기 투자가 후속 투자의 생산성을 높이는가?

Timing of Intervention

When is investment most productive?

For a terminal skill stock $\theta_{k,T}$, define the effect of investment at age or stage t as

$$M_{k,t}^{T} \equiv \frac{\partial \theta_{k,T}}{\partial I_{k,t}}.$$

- 동일한 투자라도 투입 시점에 따라 최종 능력에 미치는 효과가 달라질 수 있음
- $M_{k,t}^{T}$는 해당 시점의 투자 생산성뿐 아니라 능력의 지속성과 이후 능력 형성 과정까지 반영함
- 민감한 시기(sensitive period)는 $M_{k,t}^{T}$가 특정 발달 단계에서 특히 큰 경우를 의미함

Early intervention is not automatically optimal.

Canonical Skill-Formation Technology

$$\theta_{k,t+1} = f_{k,t}\left(\theta_{C,t}, \theta_{N,t}, I_{k,t}, \varepsilon_{k,t+1}\right), \qquad k \in \{C, N\}.$$

- $\theta_{C,t}$: cognitive skill stock
- $\theta_{N,t}$: noncognitive skill stock
- $I_{k,t}$: investment targeted at skill k
- $\varepsilon_{k,t+1}$: skill-specific shock

1. Skill Begets Skill: Self-Productivity

For $\mathsf{k} \in \{\mathsf{C}, \mathsf{N}\}$,

$$\frac{\partial \theta_{\mathsf{k},t+1}}{\partial \theta_{\mathsf{k},\mathsf{t}}} = \frac{\partial \mathsf{f}_{\mathsf{k},\mathsf{t}}}{\partial \theta_{\mathsf{k},\mathsf{t}}} > 0.$$

- Existing cognitive skills contribute to future cognitive skills.
- Existing non-cognitive skills contribute to future non-cognitive skills.
- Skill stocks persist and build upon themselves over time.

$$\theta_{\mathsf{k},\mathsf{t}} \longrightarrow \theta_{\mathsf{k},t+1} \longrightarrow \theta_{\mathsf{k},t+2}.$$

2. Cross-Skill Effects

Let $\ell \neq \mathsf{k}$ and $\mathsf{k}, \ell \in \{\mathsf{C}, \mathsf{N}\}$.

$$\frac{\partial \theta_{\mathsf{k},\mathsf{t}+1}}{\partial \theta_{\ell,\mathsf{t}}} = \frac{\partial \mathsf{f}_{\mathsf{k},\mathsf{t}}}{\partial \theta_{\ell,\mathsf{t}}}.$$

In particular,

$$\frac{\partial \theta_{\mathsf{C},\mathsf{t}+1}}{\partial \theta_{\mathsf{N},\mathsf{t}}} \quad \text{and} \quad \frac{\partial \theta_{\mathsf{N},\mathsf{t}+1}}{\partial \theta_{\mathsf{C},\mathsf{t}}}.$$

- Positive cross-effect: one skill promotes the formation of the other.
- Negative cross-effect: one skill may interfere with or substitute for the other.

3. Investment Effects

$$\frac{\partial \theta_{k,t+1}}{\partial I_{k,t}} = \frac{\partial f_{k,t}}{\partial I_{k,t}} > 0.$$

$$\frac{\partial^2 \theta_{k,t+1}}{\partial I_{k,t}^2} = \frac{\partial^2 f_{k,t}}{\partial I_{k,t}^2} \leq 0.$$

- Investment increases the production of future skills.
- The marginal productivity of investment may decrease.
- The productivity of investment may vary across developmental stages.

$$\frac{\partial f_{k,t}}{\partial I_{k,t}} \quad \text{may vary with t.}$$

Input Figure: When Should We Invest?

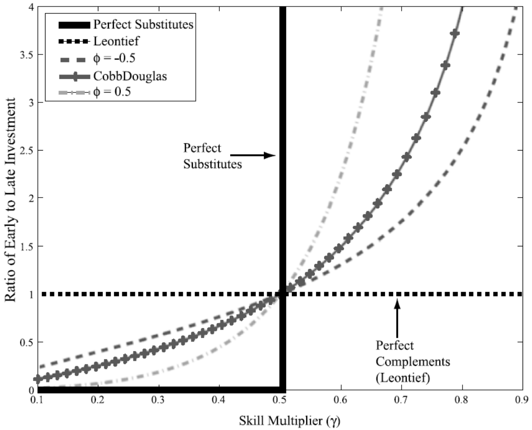

FIGURE 1.—Ratio of early to late investment in human capital as a function of the ratio of first period to second period investment productivity for different values of the complementarity parameter; assumes $r = 0$. Source: Cunha and Heckman (2007).

Why does the optimal early-to-late investment ratio depend on both γ and ϕ?

Reading the Figure

- The horizontal axis varies the relative productivity of early investment, γ.
- The vertical axis shows the optimal ratio

$$\frac{\mathsf{I}_1}{\mathsf{I}_2}.$$

- When $\gamma = 0.5$ and $\mathsf{r} = 0$, early and late investment have equal productivity. Hence,

$$\frac{\mathsf{I}_1}{\mathsf{I}_2} = 1.$$

- The shape of each curve depends on how easily early and late investments can substitute for each other.

- **Leontief:** investments are perfect complements. A balanced sequence of investment is required.
- **Cobb–Douglas:** the optimal ratio responds smoothly to relative productivity.
- **Perfect substitutes:** investment is concentrated in whichever period is more productive.

4. Direct Complementarity

$$\frac{\partial^2 \theta_{\mathsf{k},\mathsf{t}+1}}{\partial \theta_{\ell,\mathsf{t}} \partial \mathsf{I}_{\mathsf{k},\mathsf{t}}} = \frac{\partial^2 \mathsf{f}_{\mathsf{k},\mathsf{t}}}{\partial \theta_{\ell,\mathsf{t}} \partial \mathsf{I}_{\mathsf{k},\mathsf{t}}}.$$

$$\begin{cases} > 0 & \text{Direct complementarity: investment is more productive} \\ & \text{for children with higher current skills,} \\ < 0 & \text{Compensatory investment: investment is more productive} \\ & \text{for children with lower current skills.} \end{cases}$$

5. Dynamic Complementarity

Dynamic complementarity asks whether earlier investment raises the productivity of later investment:

$$\frac{\partial}{\partial \mathsf{I}_{\mathsf{t}}}\left(\frac{\partial \theta_{\mathsf{k},t+2}}{\partial \mathsf{I}_{\mathsf{k},t+1}}\right) > 0.$$

Equivalently,

$$\frac{\partial^2 \theta_{\mathsf{k},t+2}}{\partial \mathsf{I}_{\mathsf{k},t+1}\partial \mathsf{I}_{\mathsf{t}}} > 0.$$

- Early investment builds skills.
- Those skills make later investment more productive.
- This creates a dynamic mechanism through which “skills beget skills.”

A Cobb–Douglas Benchmark

Level form

$$\theta_{k,t+1} = A_{s,k}\, \theta_{C,t}^{\alpha_{s,k,C}} \theta_{N,t}^{\alpha_{s,k,N}} I_{k,t}^{\beta_{s,k}} \\ \times \theta_{C,P}^{\delta_{s,k,C}} \theta_{N,P}^{\delta_{s,k,N}} \exp(\varepsilon_{k,t+1}), \qquad k \in \{C, N\}.$$

Log form

$$\ln \theta_{k,t+1} = \ln A_{s,k} + \alpha_{s,k,C} \ln \theta_{C,t} + \alpha_{s,k,N} \ln \theta_{N,t} \\ + \beta_{s,k} \ln I_{k,t} + \delta_{s,k,C} \ln \theta_{C,P} + \delta_{s,k,N} \ln \theta_{N,P} + \varepsilon_{k,t+1}.$$

Directynamic complementarity

$$\frac{\partial^2 \ln \theta_{k,t+1}}{\partial \ln \theta_{\ell,t}\, \partial \ln I_{k,t}} = 0, \qquad \ell \in \{C, N\}.$$

With positive input elasticities, the Cobb–Douglas technology mechanically imposes dynamic complementarity:

$$\frac{\partial^2 f}{\partial \theta_{\ell,t} \partial I_{k,t}} > 0.$$

The CHS CES Specification

The specification used in Cunha, Heckman, and Schennach (2010)

$$\theta_{k,t+1} = \Big[\gamma_{s,k,1}\theta_{C,t}^{\varphi_{s,k}} + \gamma_{s,k,2}\theta_{N,t}^{\varphi_{s,k}} + \gamma_{s,k,3}I_{k,t}^{\varphi_{s,k}}$$
$$+ \gamma_{s,k,4}\theta_{C,P}^{\varphi_{s,k}} + \gamma_{s,k,5}\theta_{N,P}^{\varphi_{s,k}}\Big]^{1/\varphi_{s,k}}, \qquad k \in \{C, N\}.$$

$$\sum_{\ell=1}^{5} \gamma_{s,k,\ell} = 1, \qquad \sigma_{s,k} = \frac{1}{1-\varphi_{s,k}}.$$

- s: developmental stage
- $\gamma_{s,k,\ell}$: input-share parameters
- $\varphi_{s,k}$: curvature parameter
- $\sigma_{s,k}$: elasticity of substitution

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Cross-Partial and Directynamic Complementarity

Let $\mathsf{S}_{\mathsf{s},\mathsf{k}}$ denote the bracketed CES index in the previous slide, and let $\varphi \equiv \varphi_{\mathsf{s},\mathsf{k}}$.

$$\theta_{\mathsf{k},\mathsf{t}+1} = \mathsf{S}_{\mathsf{s},\mathsf{k}}^{1/\varphi}$$

$$\frac{\partial \theta_{\mathsf{k},\mathsf{t}+1}}{\partial \mathsf{I}_{\mathsf{k},\mathsf{t}}} = \gamma_{\mathsf{s},\mathsf{k},3} \mathsf{S}_{\mathsf{s},\mathsf{k}}^{\frac{1}{\varphi}-1} \mathsf{I}_{\mathsf{k},\mathsf{t}}^{\varphi-1}$$

For $\ell \in \{\mathsf{C}, \mathsf{N}\}$, define

$$\gamma^{\theta}_{\mathsf{s},\mathsf{k},\mathsf{C}} \equiv \gamma_{\mathsf{s},\mathsf{k},1}, \qquad \gamma^{\theta}_{\mathsf{s},\mathsf{k},\mathsf{N}} \equiv \gamma_{\mathsf{s},\mathsf{k},2}.$$

$$\frac{\partial^2 \theta_{\mathsf{k},\mathsf{t}+1}}{\partial \theta_{\ell,\mathsf{t}} \partial \mathsf{I}_{\mathsf{k},\mathsf{t}}} = (1-\varphi)\, \gamma^{\theta}_{\mathsf{s},\mathsf{k},\ell} \gamma_{\mathsf{s},\mathsf{k},3} \mathsf{S}_{\mathsf{s},\mathsf{k}}^{\frac{1}{\varphi}-2} \times \theta_{\ell,\mathsf{t}}^{\varphi-1} \mathsf{I}_{\mathsf{k},\mathsf{t}}^{\varphi-1}.$$

$$\mathsf{sign}\left(\frac{\partial^2 \theta_{\mathsf{k},\mathsf{t}+1}}{\partial \theta_{\ell,\mathsf{t}} \partial \mathsf{I}_{\mathsf{k},\mathsf{t}}}\right) = \mathsf{sign}(1-\varphi_{\mathsf{s},\mathsf{k}}).$$

Since CHS assume $\varphi_{\mathsf{s},\mathsf{k}} \leq 1$, the cross-partial is weakly positive.
This is the issue emphasized by Agostinelli and Wiswall (2025): the specification rules out negative contemporaneous complementarity, or compensatory investment for disadvantaged children.

Issues to Address

- θ is usually measured with errors
- Parental investment is naturally endogenous $\text{Cov}(\mathrm{I_t}, \varepsilon_\mathrm{t}) \neq 0$
- Initial skill itself is endogenous

Outline

Cunha, Heckman, and Schennach (2010)는 기본적으로
Simultaneous Equation Estimation Approach:

1. Production Function ($\theta_{t+1} = f(\theta_t, I_t, \varepsilon_t)$)
2. Measurement Model ($Z_{jt} = g(\lambda_{jt}, \theta, \varepsilon_{jt})$)
3. Investment Function ($I_{it} = h(X_{it}, \eta_{it})$)
4. Anchoring to the final outcome ($Y_{it} = y(\theta_T, \varepsilon_y)$)

From Skill Formation to Measurement

We do not observe skills or investments directly.

아동의 능력이나 부모의 투자를 완벽하게 관찰 할 수는 없다.

$\theta_{k,t}$, I_t are latent variables.

Instead, we observe imperfect measures:

$$Z_{k,t,j} = \mu_{k,t,j} + \alpha_{k,t,j}\theta_{k,t} + \epsilon_{k,t,j}.$$

- $Z_{k,t,j}$: observed measure j
- $\theta_{k,t}$: latent cognitive or noncognitive skill
- $\alpha_{k,t,j}$: factor loading
- $\epsilon_{k,t,j}$: measurement error

하나의 test score는 능력 그 자체가 아니라 능력의 불완전한 proxy이다.

CHS Application: Data and Developmental Stages

- Children of the NLSY/79 (CNLSY/79)
- 2,207 firstborn white children
- Assessments every two years from age 0 to 14
- Eight periods distributed over two developmental stages

Age	0	1–2	3–4	5–6	7–8	9–10	11–12	13–14
t	1	2	3	4	5	6	7	8

- Stage 1: age 0 to ages 5–6
- Stage 2: ages 5–6 to ages 13–14
- In the empirical application:

$$I_{C,t} = I_{N,t} = I_t$$

Cunha, Heckman, and Schennach (2010)

What Do These Measures Look Like?

Period	Cognitive proxy	Noncognitive proxy	Investment proxy
Birth	Weight at birth	Temperament/Difficulty Scale	Number of books
Ages 1–2	Body Parts	Temperament/Difficulty Scale	Number of books
Ages 3–4	PPVT	BPI Headstrong	How often mother reads to child
Ages 5–6 to 13–14	Reading Recognition	BPI Headstrong	How often child goes to musical shows
Maternal cognitive skill: ASVAB Arithmetic Reasoning			
Maternal noncognitive skill: Self-Esteem item “I am a failure”			

이 변수들은 latent skill이나 investment 그 자체가 아니라, CHS가 사용한 proxy이다.

The full application uses a much richer set of measures.

CHS Application: Measurement Proxies

TABLE IIB—*Continued*

	%Signal	%Noise		%Signal	%Noise
Measurements of Parental Investments			*Measurements of Parental Investments*		
Child Has a CD Player Ages 3–4	0.021	0.979	Eats With Mom/Dad Ages 11–12	0.153	0.847
How Often Child Goes on Outings Ages 5–6	0.100	0.900	How Often Child Goes to Museum Ages 11–12	0.217	0.783
Number of Books Child Has Ages 5–6	0.009	0.991	Child Has Musical Instruments Ages 11–12	0.016	0.984
How Often Mom Reads to Child Ages 5–6	0.086	0.914	Family Subscribes to Daily Newspapers Ages 11–12	0.018	0.982
How Often Child Eats With Mom/Dad Ages 5–6	0.173	0.827	Child Has Special Lessons Ages 11–12	0.013	0.987
Number of Magazines at Home Ages 5–6	0.164	0.836	How Often Child Goes to Musical Shows Ages 11–12	0.225	0.775
Child Has CD Player Ages 5–6	0.015	0.985	How Often Child Attends Family Gatherings Ages 11–12	0.103	0.897
How Often Child Goes to Museum Ages 5–6	0.296	0.704	How Often Child Is Praised Ages 11–12	0.026	0.974
Child Has Musical Instruments Ages 5–6	0.026	0.974	How Often Child Gets Positive Encouragement Ages 11–12	0.037	0.963
Family Subscribes to Daily Newspapers Ages 5–6	0.025	0.975	Number of Books Child Has Ages 13–14	0.023	0.977
Child Has Special Lessons Ages 5–6	0.020	0.980	Eats With Mom/Dad Ages 13–14	0.152	0.848
How Often Child Goes to Musical Shows Ages 5–6	0.304	0.696	How Often Child Goes to Museum Ages 13–14	0.201	0.799
How Often Child Attends Family Gatherings Ages 5–6	0.141	0.859	Child Has Musical Instruments Ages 13–14	0.015	0.985
How Often Child Is Praised Ages 5–6	0.056	0.944	Family Subscribes to Daily Newspapers Ages 13–14	0.017	0.983
How Often Child Gets Positive Encouragement Ages 5–6	0.081	0.919	Child Has Special Lessons Ages 13–14	0.012	0.988
Number of Books Child Has Ages 7–8	0.007	0.993	How Often Child Goes to Musical Shows Ages 13–14	0.224	0.776
How Often Mom Reads to Child Ages 7–8	0.113	0.887	How Often Child Attends Family Gatherings Ages 13–14	0.099	0.901
How Often Child Eats With Mom/Dad Ages 7–8	0.166	0.834	How Often Child Is Praised Ages 13–14	0.031	0.969
How Often Child Goes to Museum Ages 7–8	0.240	0.760	How Often Child Gets Positive Encouragement Ages 13–14	0.032	0.968

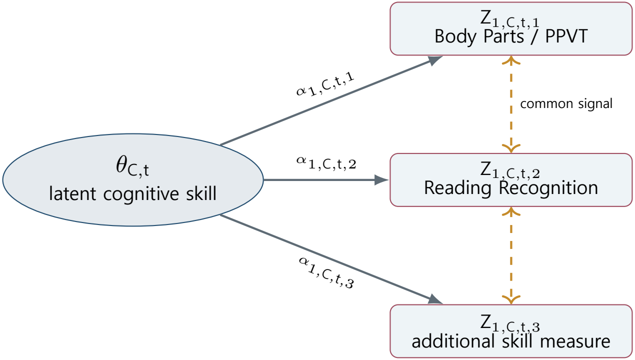
CHS Measurement Architecture

CHS divide the observed measures into several blocks.

$$
\begin{aligned}
Z_{1,k,t,j} &= \mu_{1,k,t,j} + \alpha_{1,k,t,j}\theta_{k,t} + \epsilon_{1,k,t,j}, && \text{child skill measures,} \\
Z_{2,k,t,j} &= \mu_{2,k,t,j} + \alpha_{2,k,t,j}I_{k,t} + \epsilon_{2,k,t,j}, && \text{investment measures,} \\
Z_{3,k,1,j} &= \mu_{3,k,1,j} + \alpha_{3,k,1,j}\theta_{k,P} + \epsilon_{3,k,1,j}, && \text{parental skill measures.}
\end{aligned}
$$

- $k \in \{C, N\}$: cognitive and noncognitive dimensions
- j: different observed measures within each block
- $\theta_{k,t}$, $I_{k,t}$, and $\theta_{k,P}$ are latent

CHS Measurement: Cognitive Skill Indicators

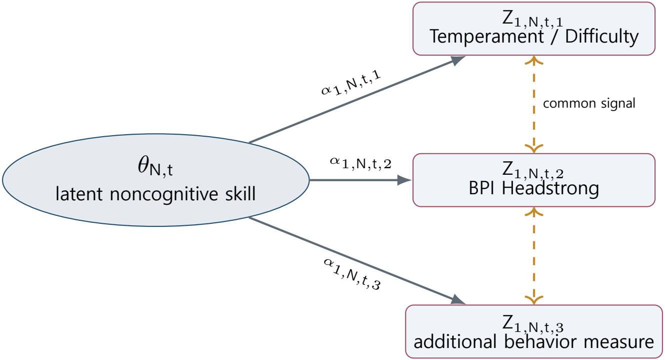


$$Z_{1,C,t,j} = \mu_{1,C,t,j} + \alpha_{1,C,t,j}\theta_{C,t} + \epsilon_{1,C,t,j}.$$

$$\text{Cov}\left(Z_{1,C,t,j}, Z_{1,C,t,m}\right) = \alpha_{1,C,t,j}\alpha_{1,C,t,m}\,\text{Var}(\theta_{C,t}) \neq 0.$$

- The observed measures are correlated through their common latent skill.
- They are not mutually exclusive indicators of different children or different processes.

CHS Measurement: Noncognitive Skill Indicators



$$Z_{1,N,t,j} = \mu_{1,N,t,j} + \alpha_{1,N,t,j}\theta_{N,t} + \epsilon_{1,N,t,j}.$$

$$\mathrm{Cov}\left(Z_{1,N,t,j}, Z_{1,N,t,m}\right) = \alpha_{1,N,t,j}\alpha_{1,N,t,m}\,\mathrm{Var}(\theta_{N,t}) \neq 0.$$

- Each proxy has its own factor loading.
- The correlation among measures helps identify the common latent factor.

Step 1: Use Multiple Measures to Recover Latent Factors

$$
\begin{aligned}
Z_{1,t,1} &= \mu_{1,t,1} + \alpha_{1,t,1}\theta_t + \epsilon_{1,t,1}, \\
Z_{1,t,2} &= \mu_{1,t,2} + \alpha_{1,t,2}\theta_t + \epsilon_{1,t,2}, \\
Z_{1,t,3} &= \mu_{1,t,3} + \alpha_{1,t,3}\theta_t + \epsilon_{1,t,3}.
\end{aligned}
$$

$$
\mathrm{Cov}(Z_{1,t,j}, Z_{1,t,m})
$$

contains information about the common latent component θ_t.

- Common covariance across measures identifies the latent signal.
- Idiosyncratic variation is assigned to measurement error.
- Factor loadings are estimated rather than choosing arbitrary weights.

예를 들어 Body Parts, PPVT, Reading Recognition은 서로 다른 관측치이지만 cognitive skill이라는 공통 요인을 측정한다.

Step 1A: Covariance Reveals the Common Signal

For simplicity, suppress the factor index and write

$$Z_{t,j} = \mu_{t,j} + \alpha_{t,j}\theta_t + \epsilon_{t,j}.$$

Assume that measurement errors are orthogonal to the latent factor and uncorrelated across measures:

$$\mathrm{Cov}(\theta_t, \epsilon_{t,j}) = 0, \qquad \mathrm{Cov}(\epsilon_{t,j}, \epsilon_{t,m}) = 0 \quad (j \neq m).$$

Then, for two different measures,

$$\begin{aligned} \mathrm{Cov}(Z_{t,j}, Z_{t,m}) &= \mathrm{Cov}\left(\alpha_{t,j}\theta_t + \epsilon_{t,j}, \alpha_{t,m}\theta_t + \epsilon_{t,m}\right) \\ &= \alpha_{t,j}\alpha_{t,m}\,\mathrm{Var}(\theta_t). \end{aligned}$$

$$\boxed{\mathrm{Cov}(Z_{t,j}, Z_{t,m}) = \underbrace{\alpha_{t,j}\alpha_{t,m}\,\mathrm{Var}(\theta_t)}_{\text{common latent signal}}}$$

- Covariance across measures reflects the common latent factor.
- Idiosyncratic measurement error disappears from cross-measure covariance.
- Before normalization, the factor is identified only up to location and scale.

Step 2: Fix the CHS Factor Scale

Latent factors have no natural units.

$$\theta_t \quad \text{and} \quad a + b\theta_t$$

can generate observationally equivalent measurement equations unless we impose normalizations.

CHS: Period-by-Period Normalization

$$\boxed{\alpha_{a,k,t,1} = 1 \qquad \text{for every } t}$$

$$\boxed{E(\theta_{k,t}) = 0, \qquad E(I_{k,t}) = 0 \qquad \text{for every } t}$$

- The first loading fixes the scale separately at each period.
- Mean-zero restrictions recenter each period's latent factor.

Agostinelli–Wiswall의 시점 간 정규화와 식별 문제는 다음 강의에서 다룬다.

Step 3: Anchor Skills in Adult Outcomes

Test scores are useful proxies, but their units are arbitrary.
CHS use adult outcomes to give the latent skills an interpretable scale.

$$Z_{4,j} = \mu_{4,j} + \alpha_{4,C,j}\theta_{C,T+1} + \alpha_{4,N,j}\theta_{N,T+1} + \epsilon_{4,j}.$$

For a linear adult outcome, such as years of schooling,

$$g_{C,j}(\theta_{C,T+1}) = \mu_{4,j} + \alpha_{4,C,j}\theta_{C,T+1},$$

$$g_{N,j}(\theta_{N,T+1}) = \mu_{4,j} + \alpha_{4,N,j}\theta_{N,T+1}.$$

- Adult outcomes provide an economically interpretable metric.
- The same latent skill can be compared across childhood periods.
- CHS use educational attainment as the main linear anchor in their empirical tables.

Step 3A: Identifying the Adult-Outcome Loading

For simplicity, focus on the noncognitive factor and suppress the cognitive factor.

$$\mathsf{Z}_{4,\mathsf{j}} = \mu_{4,\mathsf{j}} + \alpha_{4,\mathsf{N},\mathsf{j}}\theta_{\mathsf{N},\mathsf{T}+1} + \epsilon_{4,\mathsf{j}}.$$

Assume

$$\mathsf{Cov}\left(\theta_{\mathsf{N},\mathsf{T}+1}, \epsilon_{4,\mathsf{j}}\right) = 0.$$

Taking covariance with the latent skill gives

$$\begin{aligned}\mathsf{Cov}\left(\mathsf{Z}_{4,\mathsf{j}}, \theta_{\mathsf{N},\mathsf{T}+1}\right) &= \mathsf{Cov}\left(\alpha_{4,\mathsf{N},\mathsf{j}}\theta_{\mathsf{N},\mathsf{T}+1} + \epsilon_{4,\mathsf{j}}, \theta_{\mathsf{N},\mathsf{T}+1}\right) \\ &= \alpha_{4,\mathsf{N},\mathsf{j}}\,\mathsf{Var}\left(\theta_{\mathsf{N},\mathsf{T}+1}\right).\end{aligned}$$

Therefore,

$$\boxed{\alpha_{4,\mathsf{N},\mathsf{j}} = \frac{\mathsf{Cov}\left(\mathsf{Z}_{4,\mathsf{j}}, \theta_{\mathsf{N},\mathsf{T}+1}\right)}{\mathsf{Var}\left(\theta_{\mathsf{N},\mathsf{T}+1}\right)}}$$

인지능력도 함께 포함하는 경우에는 인지능력을 통제한 partial covariance로 해석한다.

Step 3B: Where Does the Factor Variance Come From?

Under uncorrelated measurement errors,

$$
\begin{aligned}
\mathsf{C}_{12} &\equiv \mathsf{Cov}(\mathsf{Z}_{\mathsf{N},1}, \mathsf{Z}_{\mathsf{N},2}) = \lambda_{\mathsf{N},1}\lambda_{\mathsf{N},2}\mathsf{V}_{\mathsf{N}}, \\
\mathsf{C}_{13} &= \lambda_{\mathsf{N},1}\lambda_{\mathsf{N},3}\mathsf{V}_{\mathsf{N}}, \\
\mathsf{C}_{23} &= \lambda_{\mathsf{N},2}\lambda_{\mathsf{N},3}\mathsf{V}_{\mathsf{N}}.
\end{aligned}
$$

Using the normalization $\lambda_{\mathsf{N},1} = 1$,

$$
\boxed{\mathsf{Var}(\theta_{\mathsf{N},\mathsf{T}+1}) = \frac{\mathsf{C}_{12}\mathsf{C}_{13}}{\mathsf{C}_{23}}}
$$

Step 3B: Continued

Thus, the denominator in the adult-outcome loading contains information from the covariance structure of the other measures.

$$\boxed{\alpha_{4,\mathsf{N},\mathsf{j}} = \frac{\mathsf{Cov}(\mathsf{Z}_{4,\mathsf{j}}, \mathsf{Z}_{\mathsf{N},1})}{\mathsf{Var}(\theta_{\mathsf{N},\mathsf{T}+1})} = \frac{\mathsf{Cov}(\mathsf{Z}_{4,\mathsf{j}}, \mathsf{Z}_{\mathsf{N},1})\mathsf{C}_{23}}{\mathsf{C}_{12}\mathsf{C}_{13}}}$$

- $\mathsf{Z}_{\mathsf{N},1}$ fixes the factor's unit.
- $\mathsf{Z}_{\mathsf{N},2}$ and $\mathsf{Z}_{\mathsf{N},3}$ help identify the latent variance.
- The remaining measures need not be normalized individually.

What Outcomes Do Papers Use?

Cunha, Heckman, and Schennach (2010)

- **Main empirical anchor:** completed years of education by age 19
- **Child skill indicators:** cognitive test scores and personality or behavioral assessments
- The broader framework allows schooling, wages, occupational attainment, hours worked, criminal activity, and teenage pregnancy as outcomes

Earlier Heckman–Cunha applications

- **Cunha and Heckman (2007):** human capital as a composite adult outcome; policy outcomes include high-school graduation, college enrollment, conviction, probation, and welfare use
- **Heckman, Stixrud, and Urzua (2006):** schooling, wages, employment, work experience, occupational choice, teenage pregnancy, marriage, smoking, marijuana use, and illegal activities

Step 4: Connect Measurement to Production

The measurement system produces estimates of the latent variables:

$$\widehat{\theta}_{C,t}, \quad \widehat{\theta}_{N,t}, \quad \widehat{I}_t, \quad \widehat{\theta}_{C,P}, \quad \widehat{\theta}_{N,P}.$$

These latent variables enter the skill-formation technology:

$$\theta_{k,t+1} = \left[\sum_{\ell=1}^{5} \gamma_{s,k,\ell} X_{\ell,t}^{\varphi_{s,k}}\right]^{1/\varphi_{s,k}} \exp(\eta_{k,t+1}).$$

- Measurement equations describe how latent factors generate observed data.
- The production function describes how latent skills and investment evolve.
- CHS estimate the system using parametric maximum likelihood.
- The latent factors are integrated out of the likelihood.

CHS: Identification versus Estimation

Identification

CHS show that the joint distribution of latent skills, investments, and parental endowments can be identified from multiple measurements, under appropriate independence and rank conditions.

Empirical estimation

For the application, CHS impose parametric assumptions and estimate the complete system by maximum likelihood.

$$\widehat{\vartheta} = \arg\max_{\vartheta} \sum_{i=1}^{N} \log L_i(\vartheta).$$

- Nonparametric identification is a theoretical result.
- The reported estimates come from a parametric likelihood.
- The likelihood incorporates measurement equations, production functions, investment equations, and adult outcomes.

Is the Factor Model Solved in Closed Form?

Simplified measurement model

With a linear factor model and strong assumptions, some loading parameters can be obtained from covariance ratios:

$$\lambda_2 = \frac{\mathrm{Cov}(Z_1, Z_2)}{\mathrm{Var}(\theta)}.$$

The complete model does not have a closed-form solution because it contains:

- nonlinear CES skill production;
- dynamic latent states;
- endogenous investments;
- missing observations and adult-outcome anchoring.

The factor model is identified analytically in simplified cases, but estimated numerically in the full CHS system.

Signal and Noise in Observed Measures

For one skill dimension $k \in \{C, N\}$,

$$Z_{a,k,t,j} = \mu_{a,k,t,j} + \underbrace{\alpha_{a,k,t,j}\theta_{k,t}}_{\text{common signal}} + \underbrace{\epsilon_{a,k,t,j}}_{\text{measure-specific noise}}.$$

Observed measure	**Common signal**	**Measure-specific noise**
Cognitive tests (PPVT, Reading Recognition, Body Parts)	$\alpha_{C,t,j}\theta_{C,t}$	문항 난이도, 당일 집중도, 검사 환경, test-specific scoring error
Noncognitive measures (behavioral and emotional assessments)	$\alpha_{N,t,j}\theta_{N,t}$	보고자 특성, 상황 특수적 행동, 응답오차 및 일시적 상태
Different measures of the same skill	동일한 latent factor를 공유	각 측정치에만 존재하는 idiosyncratic variation

Key interpretation

An observed score is not the latent skill itself.

$$\text{Observed measure} = \text{latent-skill signal} + \text{measurement noise}.$$

높은 점수가 항상 높은 능력을 의미하는 것은 아니다.

CHS Measurement Quality: Cognitive Proxies

For each observed measure,

$$Z_{C,t,j} = \mu_{C,t,j} + \alpha_{C,t,j} \ln \theta_{C,t} + \epsilon_{C,t,j}.$$

$$s^{\theta}_{C,t,j} = \frac{\alpha^2_{C,t,j}\,\mathrm{Var}(\ln \theta_{C,t})}{\alpha^2_{C,t,j}\,\mathrm{Var}(\ln \theta_{C,t}) + \mathrm{Var}(\epsilon_{C,t,j})}, \qquad s^{\epsilon}_{C,t,j} = 1 - s^{\theta}_{C,t,j}.$$

Age	Cognitive measure	% Signal	% Noise
Birth	Motor–Social Development	4.5	95.5
Birth	Weight at Birth	55.7	44.3
Ages 1–2	Body Parts	30.8	69.2
Ages 1–2	Memory for Locations	16.0	84.0
Ages 3–4	Picture Vocabulary (PPVT)	43.1	56.9
Ages 5–6	PIAT Mathematics	31.4	68.6
Ages 5–6	PIAT Reading Recognition	95.8	4.2
Ages 7–8	PIAT Reading Recognition	86.9	13.1
Ages 13–14	PIAT Reading Recognition	73.5	26.5

- 동일한 cognitive skill을 측정하더라도 측정치에 따라 정보량이 크게 다르다.
- Ages 5–6에서 Reading Recognition은 95.8%가 signal이지만, PIAT Mathematics는 31.4%에 불과하다.
- CHS는 전체 cognitive measures에서 약 54%가 latent skill signal이라고 보고한다.

CHS Measurement Quality: Noncognitive Proxies

The same decomposition applies to noncognitive measures:

$$Z_{N,t,j} = \mu_{N,t,j} + \alpha_{N,t,j} \ln \theta_{N,t} + \epsilon_{N,t,j}.$$

Age	Noncognitive measure	% Signal	% Noise
Birth	Difficulty	15.1	84.9
Ages 1–2	Difficulty	38.2	61.8
Ages 1–2	Sociability	7.5	92.5
Ages 3–4	Sociability	0.8	99.2
Ages 3–4	BPI Headstrong	51.8	48.2
Ages 5–6	BPI Headstrong	61.1	38.9
Ages 7–8	BPI Headstrong	60.5	39.5
Ages 11–12	BPI Headstrong	60.3	39.7
Ages 13–14	BPI Headstrong	59.5	40.5

- Noncognitive measures는 cognitive measures보다 전반적으로 정보량이 낮다.
- 같은 시기에도 Sociability는 0.8%만 signal인 반면, BPI Headstrong은 51.8%가 signal이다.
- CHS는 전체 childhood noncognitive measures에서 latent factor가 설명하는 비율이 40% 미만이라고 보고한다.
- 따라서 관측된 noncognitive score의 상당 부분은 measurement noise이다.

Estimates: Cognitive Skill Formation

	First Stage		Second Stage	
	Without ME	**With ME**	Without ME	**With ME**
Self-productivity $\gamma_{s,C,1}$	0.403 (0.058)	0.487 (0.030)	0.657 (0.013)	0.902 (0.014)
Cross-productivity $\gamma_{s,C,2}$	0.218 (0.105)	0.083 (0.026)	0.009 (0.005)	0.011 (0.005)
Current investments $\gamma_{s,C,3}$	0.067 (0.090)	0.231 (0.024)	0.167 (0.018)	0.020 (0.006)
Parental cognitive skills $\gamma_{s,C,4}$	0.268 (0.078)	0.050 (0.013)	0.047 (0.009)	0.047 (0.008)
Parental noncognitive skills $\gamma_{s,C,5}$	0.044 (0.050)	0.148 (0.030)	0.119 (0.150)	0.020 (0.010)
Complementarity parameter $\varphi_{s,C}$	0.375 (0.294)	0.611 (0.240)	-0.827 (0.093)	-1.373 (0.168)
Elasticity of substitution $\sigma_{s,C}$	1.601	2.569	0.547	0.421
Variance of shocks $\delta^2_{s,C}$	0.941 (0.048)	0.165 (0.007)	0.358 (0.006)	0.097 (0.003)

Cognitive Skills

Main comparison

- The estimated early investment effect increases:

$$\gamma_{1,\mathrm{C},3}: \quad 0.067 \longrightarrow 0.231$$

- The estimated late investment effect decreases:

$$\gamma_{2,\mathrm{C},3}: \quad 0.167 \longrightarrow 0.020$$

- Measurement error makes early investment look less important and late investment look more important.
- Cognitive self-productivity becomes substantially stronger, especially in the second stage.

Without ME: CHS Table III. With ME: CHS Table I. Standard errors are reported in parentheses.

Estimates: Noncognitive Skill Formation

	First Stage		Second Stage	
	Without ME	With ME	Without ME	With ME
Cross-productivity $\gamma_{s,N,1}$	0.193 (0.095)	0.000 (0.025)	0.058 (0.014)	0.008 (0.010)
Self-productivity $\gamma_{s,N,2}$	0.594 (0.090)	0.649 (0.034)	0.638 (0.020)	0.868 (0.011)
Current investments $\gamma_{s,N,3}$	0.099 (0.296)	0.146 (0.027)	0.239 (0.031)	0.055 (0.013)
Parental cognitive skills $\gamma_{s,N,4}$	0.114 (0.055)	0.022 (0.011)	0.065 (0.015)	0.000 (0.007)
Parental noncognitive skills $\gamma_{s,N,5}$	0.000 (0.821)	0.183 (0.031)	0.000 (0.203)	0.069 (0.017)
Complementarity parameter $\varphi_{s,N}$	-0.723 (0.441)	-0.674 (0.324)	-0.716 (0.127)	-0.695 (0.274)
Elasticity of substitution $\sigma_{s,N}$	0.580	0.597	0.583	0.590
Variance of shocks $\delta^2_{s,N}$	0.767 (0.076)	0.189 (0.012)	0.597 (0.017)	0.103 (0.004)

Non-cognitive skill formation

Main comparison

- The apparent cognitive-to-noncognitive effect disappears:

$$\gamma_{1,\mathrm{N},1}: \quad 0.193 \longrightarrow 0.000$$

- Noncognitive self-productivity becomes stronger, especially in the second stage:

$$\gamma_{2,\mathrm{N},2}: \quad 0.638 \longrightarrow 0.868$$

- The estimated early investment effect increases, while the estimated late investment effect decreases.
- Parental noncognitive skills become important for the formation of child noncognitive skills.

Without ME: CHS Table III. With ME: CHS Table I. Standard errors are reported in parentheses.

Investment Function

1. Skill formation technology and investment choice are distinct objects

$$\theta_{k,t+1} = f_k(\theta_t, I_t, \theta_P, \varepsilon_{k,t+1})$$

$$I_t = \pi(X_t, \theta_t, \theta_P, p_t, y_t, \nu_t)$$

2. Modeling $\pi(\cdot)$ as a full structural choice rule requires specifying parents' preferences, constraints, information, prices, and expectations.
3. Therefore, many empirical applications focus on estimating $f_k(\cdot)$ and treat investment as an observed input.
4. When investment is potentially endogenous, a common alternative is to estimate a reduced-form investment equation and use a control function:

$$I_t = \pi(X_t, Z_t) + \nu_t, \qquad \widehat{\nu}_t \text{ is included in the skill-formation equation.}$$

Attanasio et al. (2020) use a control-function approach to account for potentially endogenous parental investments.

Additional Materials Not Covered

Cunha, Heckman, and Schennach (2010) develop a broader framework than the material covered in this lecture.

- **Nonparametric identification of the production technology**
  - Identification of a broad class of nonlinear technologies with measurement error
- **Nonlinear anchoring using adult outcomes**
  - High-school graduation, criminal activity, drug use, and teenage pregnancy
- **Endogenous parental investments**
  - Time-invariant and time-varying unobserved heterogeneity
  - Investment policy functions and exclusion restrictions
- **Nonnormal latent-factor distributions**
  - Mixture-of-normals specifications
- **Optimal investment and policy counterfactuals**
  - Simulating optimal early and late investments under alternative initial conditions

Scope of this lecture

We focus on the measurement system and the parametric CES skill-formation technology.

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